>

>

>

> *restart*;

> *with*(*Statistics*) :

>

```
> with(stats):
> with(stats[statplots]):
```

>

> *#Binomial distribution*

> $Pm := \frac{n!}{m! \cdot (n-m)!} \cdot p^m \cdot q^{(n-m)}$;

$$Pm := \frac{n!\, p^m\, q^{n-m}}{m!\, (n-m)!} \quad \textbf{(1)}$$

> $q := 1 - p$;

$$q := 1 - p \quad \textbf{(2)}$$

> $n := 3$; $p := 0.5$;

$$n := 3$$
$$p := 0.5 \quad \textbf{(3)}$$

> **for** *m* **from** 0 **to** *n* **do**
 $P[m] := Pm$:
 $dd[m+1] := Weight(m..m+1, P[m])$:
 end do:

>

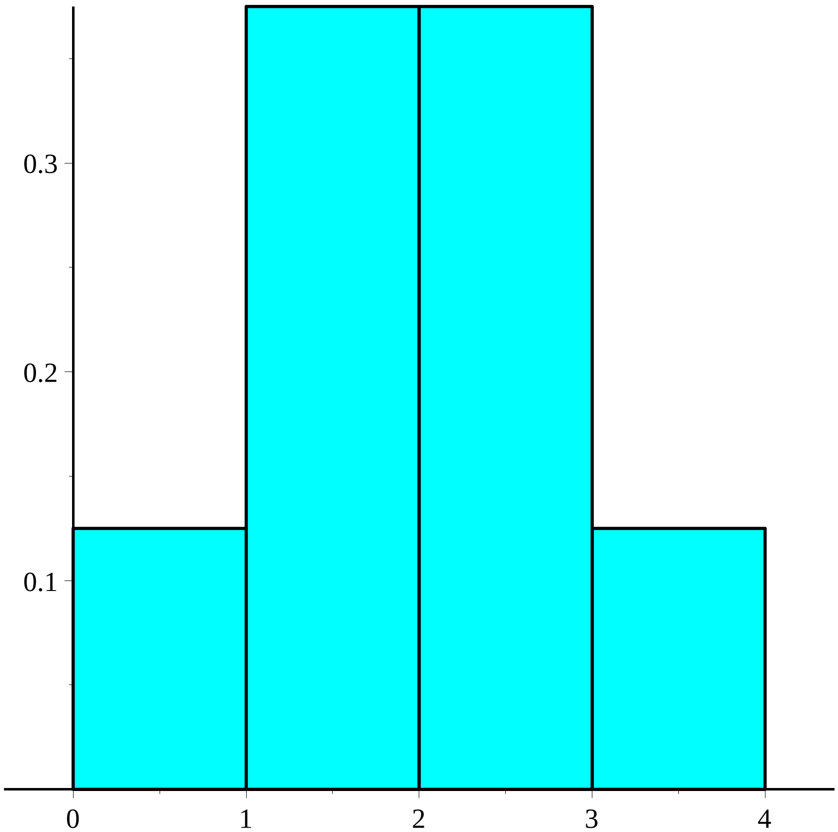
> **for** m **from** 0 **to** *n* **do**
 printf ("%f %f \n", *m*, *evalf* (*P*[*m*])) :
 end do;

```
0.000000  0.125000
1.000000  0.375000
2.000000  0.375000
3.000000  0.125000
```

>

> $dP := [seq(dd[i], i = 1..n+1)]$:

> *histogram*(*dP*, *color* = *cyan*);



> $\mu x := 0;$

$$\mu x := 0 \quad (4)$$

> **for** m **from** 0 **to** n **do**
$\mu x := \mu x + m \cdot P[m]:$
end do:

> $print(\mu x= , \mu x);$

$$\mu x=, 1.5000 \quad (5)$$

> $Dx := 0;$

$$Dx := 0 \quad (6)$$

> **for** m **from** 0 **to** n **do**
$Dx := Dx + (m - \mu x)^2 \cdot P[m]:$
end do:

> $print(`Dx=`, Dx);$

$$Dx=, 0.7500000000 \quad (7)$$

> $\sigma := \mathrm{sqrt}(Dx);$

$$\sigma := 0.8660254038 \quad (8)$$

>